sphere; **(**b**)** *zero* curvature, which does not bloat at all, as on an infinite plane or finite video game screen; **(**c**)** *negative* curvature, which uniformly shrinks inward, as on a saddle.

As input, Einstein's equations take the amount of matter and energy in the universe (assumed, again by consideration of symmetry, to be distributed uniformly) and as output, they give the curvature of space. The difficulty is that for many decades astronomers have been unable to agree on how much matter and energy there actually is. If all the matter and energy in the universe were to be smeared uniformly throughout space, and if, after this was done, there turned out to be more than the so-called critical density of about .00000000000000000001 (10^{-23}) grams in every cubic meter[22]—about five hydrogen atoms per cubic meter—Einstein's equations would yield a positive curvature for space; if there were less than the critical density, the equations would imply negative curvature; if there were exactly the critical density, the equations would tell us that space has no overall curvature. Although this observational issue is yet to be settled definitively, the most refined data are tipping the scales on the side of no curvature—the flat shape. But the question of whether the Energizer Bunny could move forever in one direction and vanish into the darkness, or would one day circle around and catch you from behind—whether space goes on forever or wraps back like a video screen—is still completely open.[14]

Even so, even without a final answer to the shape of the cosmic fabric, what's abundantly clear is that symmetry is *the* essential consideration allowing us to comprehend space and time when applied to the universe as a whole. Without invoking the power of symmetry, we'd be stuck at square one.

Cosmology and Spacetime

We can now illustrate cosmic history by combining the concept of expanding space with the loaf-of-bread description of spacetime from Chapter 3. Remember, in the loaf-of-bread portrayal, each slice—even though two-dimensional—represents all of three-dimensional space at a single moment of time from the perspective of one particular observer.